The Canadian Urban Environmental Health Research Consortium

Canue Metadata - Air Quality Air Pollution Index

2026-11-06

DATA SET INFORMATION

Dataset Code: AQAIX_AVA_YY

Description:

CANUE specialists developed multi-pollutant indices of air quality. Three-year (2012-14) average datasets of trace gases and aerosols derived from satellite-based sensors were prepared for a common 10x10 km grid over North America (NA: Canada and the United States), including surface PM2.5 (in ??g/m3) , NO2, SO2, CO and NH3 (all in ppb), and vertical column totals for HCHO (formaldehyde in molecules per cm2). Surface O3 concentrations from CANUE holdings (warm season average 8 hr max) were merged with these satellite-based data. Principal Component Analysis (PCA) was used to compute three pollutant mixture indicators for each 10x10 km grid. These values were normalized to a 0-100 scale. Each indicator is related to a different aspect of common air pollution mixtures occurring across North America (Canada and the U.S.).Indicator 1 (PC1) reflects combustion-related pollutant mixtures, typically linked to populated areas. PC1 is suggested for use as the primary multi-pollutant, long-term outdoor air pollution exposure index because it incorporates the largest diversity of pollutants (PM2.5, CO, NO2, formaldehyde (HCHO), SO2, NH3 and O3, listed here in order of their PCA loading coefficients (e.g., weights)).Indicator 2 (PC2) is related to O3 NH3 mixtures common in parts of western NA linked to higher elevations (i.e., background O3) and livestock grazing. Indicator 3 (PC3) is related to NH3 mixtures (e.g., with some weight from CO and PM2.5) associated with more intensive agricultural practices (e.g., crops, intensive raising of animals).Also included are the input air quality datasets.CANUE staff indexed the data to DMTI Spatial single-link postal codes.

Keywords: air quality|multi-pollutant|pm2.5|no2|so2|o3|co|nh3

Place Keywords: Canada|national

GEOSPATIAL REFERENCE

Upper Left Corner: 65.14N , -141.02W

Lower Right Corner: 41.68N , -52.62W

Coordinate System: GCS_WGS84 - EPSG:4326

Geometry Type: POINT - Units: Decimal Degree

Geometry Data Source: DMTI Spatial Inc. (postal codes)

QUALITY ASSESSMENT

QA/QC Procedures:

CANUE did not assess the quality of the input data or indices. Users should review the supporting documentation and any recommended citations.

Geographic Coordinate Positional Accuracy:

These metrics are linked to the corresponding annual postal codes files for mapping and analysis purposes. Refer to the postal code metadatafile in Supporting Documentation for more information.

Vertical Positional Accuracy:

Attribute Accuracy:

Data Validity: NoData = -9999 (for numeric fields) - NoData=null (for category fields) - Data insufficient to calculate value = -1111

Associated Files:

Data Comment:

DATA SOURCE

Data Source

Spatial Resolution:

Data Preparation Date: January 28, 2022

Beginning Date: 2012

End Date: 2014

Sampling Frequency of Data:

Years Available:
2012 - 2013 - 2014

MAINTENANCE

Description:

File Type: Comma separated values(.csv)

File Size:

Number of Data Files: 1

DATA USE CONDITIONS

The Data User is REQUIRED:
(i) to acknowledge data sources listed under Acknowledgement(s)
(ii) cite the publication(s) listed under References as the providers and source of these data when using them in support of research, analysis, operations, policy decision or any other undertaking including publication
(iii) complete and sign the CANUE Data Use and Sharing Agreement in which the name and signature of the researcher/analyst who takes responsibility for ensuring all conditions are met.

Data Sharing Restrictions:
These data files are provided solely for the purposes stated in the CANUE Data Sharing and Use Agreement and should not be re-distributed for any reason. These data also contain proprietary postal code data and may only be used for the project named in the CANUE Data Sharing and Use Agreement. Data can be shared only within a project team for the exclusive purposes of teaching, academic research and publishing, and/or planning of educational services in accordance to DMTI End User Agreement associated with the Spatial Mapping Academic Research Tools (SMART) Program.

Include the following references in any publications resulting from the use of these data:
Please contact CANUE for citation.

Include the following acknowledgements:
Multi-pollutant indices and input air quality data, indexed to DMTI Spatial Inc. postal codes , were provided by CANUE (Canadian Urban Environmental Health ResearchConsortium).

SUPPORT DOCUMENTATION

1 - PCA readme file (https://canuedata.ca/docs/Read_Me_PCA.docx)

VARIABLES

AQAIXYY_01 - Index - Principal Component 1

Indicator 1 scaled 0 to 100

AQAIXYY_02 - Index - Principal Component 2

Indicator 2 scaled 0 to 100

AQAIXYY_03 - Index - Principal Component 3

Indicator 3 scaled 0 to 100

AQAIXYY_04 - CO

Annual Mean MOPITT CO (carbon monoxide) Surface Concentration (2012-2014) in ppb

AQAIXYY_05 - HCHO

Annual Mean OMI HCH (formaldehyde) tropospheric vertical column density (2012-2014) in molecules per cm2

AQAIXYY_06 - NH3

Annual Mean CrIS Satellite NH3 (ammonia) Surface Concentration (2012-2014) in ppb

AQAIXYY_07 - NO2

Annual Mean OMI Satellite-derived NO2 (nitrogen dioxide) Surface Concentration (2012-2014) in ppb

AQAIXYY_08 - O3

Warm Season Average Daily 8 Hour Maximum Surface Concentration (2013-2015) in ppb

AQAIXYY_09 - PM2.5

Annual Mean Satellite-derived PM2.5 (fine particulates) Surface Concentration (2012-2014) in ug/m3

AQAIXYY_10 - SO2

Annual Mean OMI Satellite-derived SO2 (sulfur dioxide) Surface Concentration (2012-2014) in ppb

SUPPORT CONTACT

Data Set Support Contact: info@canue.ca

Affiliated Organization:
CANUE (Canadian Urban Environmental Health Research Consortium)
Dalla Lana School of Public Health, University of Toronto

WebSite: https://www.canue.ca

Toronto - Ontario - Canada

DATA SOURCE CONTACT

Data Set Support Contact: Shailesh Kumar Kharol

Email: shailesh.kharol@canada.ca; shailesh.kharol@gmail.com

Affiliated Organization:
CANUE

Toronto - Ontario - Canada